Section 1.6 — Composition of Functions — Answer Key

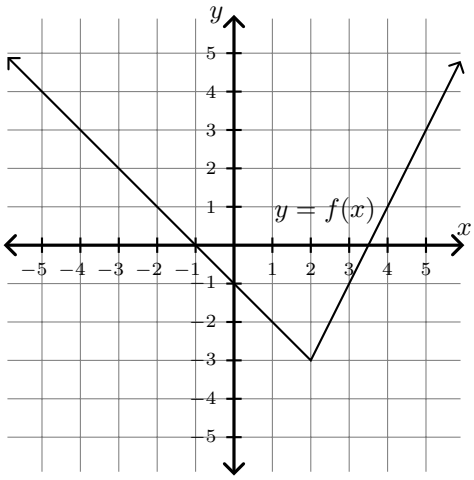
Math 103: College Algebra

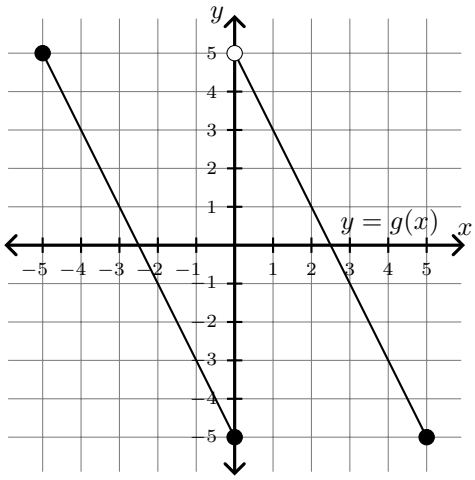
1 Composing Functions

Let $f = \{(1,5),(2,3),(3,1),(4,4),(5,2)\}$ and $g(x) = \{(1,1),(2,4),(3,2),(4,5),(5,3)\}$. Find each of the following.

1. $(f \circ g)(4) = 2$
2. $(g \circ f)(2) = 2$
3. $(f \circ g)(1) = 5$
4. $(f \circ g \circ f)(3) = 5$
5. $(f \circ f)(5) = 3$
6. $f \circ g = \big\{(1,5),(2,4),(3,3),(5,2)\big\}$

Let $f(x)$ and $g(x)$ be the functions graphed below.





Find each of the following.

7. $(f \circ g)(-1) = 2$
8. $(g \circ f)(-1) = -5$
9. $(g \circ f)(3) - 3$
10. $(f \circ f)(-2) - 2$

Let $f(x) = 2x - 4$, $g(x) = -\frac{2}{3}x + 1$, and $h(x) = |x + 2|$. Find each of the following.

11. $(f \circ g)(12) = 18$
12. $(h \circ g)(0) = 3$
13. $(h \circ f)(-1) = 4$
14. $(f \circ f)(1) = 8$
15. $(g \circ h)(1) = -1$
16. $(f \circ g)(x) = -\frac{4}{3}x - 2$
17. $(g \circ h \circ f)(x) = -\frac{2}{3}\,|2x - 4 + 2| + 1$

2 Inverses

Let $f(x) = \frac{1}{2}x - 4$, $g(x) = -2x + 4$, $h(x) = -\frac{1}{2}x + 2$, $k(x) = 2x + 2$. Give expressions for each of the following:

18. $(f \circ g)(x) = -x - 2$
19. $(f \circ h)(x) - \frac{1}{4}x - 3$
20. $(f \circ k)(x) = x - 3$
21. $(g \circ h)(x) = x$
22. $(g \circ k)(x) = -4x$
23. $(h \circ k)(x) = -x + 1$
24. g and h are inverses of one another.

3 Undoing composition

Give functions so that the given function is the given composition. Each function should only do one thing.

25. $f(x) = x - 4$, $g(x) = 2x$
26. $f(x) = |x|$, $g(x) = x + 1$
27. $f(x) = x + 4$, $g(x) = x^2$
28. $f(x) = x^4$, $g(x) = x - 3$
29. $f(x) = 3x$, $g(x) = |x|$
30. $f(x) = \sqrt{x}$, $g(x) = x - 5$

4 Domain and Range

For each pair of functions f and g, find the domain and range of $f \circ g$.

31. $\operatorname{Dom} f \circ g = \{3, 9\}$
    $\operatorname{Rng} f \circ g = \{4\}$

32. $\operatorname{Dom} f \circ g = \{-7, 3, 9\}$
    $\operatorname{Rng} f \circ g = \{1\}$

33. $\operatorname{Dom} f \circ g = \varnothing$
    $\operatorname{Rng} f \circ g = \varnothing$

34. $\operatorname{Dom} f \circ g = \{-7, -2\}$
    $\operatorname{Rng} f \circ g = \{-7, 8\}$

For each pair of functions f and g, find the domain of $f \circ g$.

35. $\operatorname{Dom} f \circ g = (0, \infty)$

36. $\operatorname{Dom} f \circ g = \left[0, \frac{5}{2}\right]$